

Multiple Results of Startup Predictive Analysis using 1. Multiple Linear Regression , 2. SVM & 3. Decision Tree Regression

1. Multiple Linear Regression :

Regressor: LinearRegression() (Default Parameters)

R2 Score : 0.9358680970046243 **Predicted Result:** 44586.30

2. SVM (Support Vector Machine) Regression :

Regressor:

Parameters Tweaked are kernel and C - SVR(kernel = " ", C = " ")

kernel Values are : "linear", "poly", "rbf" & "sigmoid"

c_vales are : 0.025, 0.05, 0.1, 0.5, 1.0, 1.5

R2 Score :

For C =0.025, R_Square Scores: r_score_linear = 0.9355209342784299 r_score_Poly = -0.05732080221260971 r_score_RBF = -0.05748206464442651 r_score_Sigmoid = -0.057486613037766476	For C =0.5, R_Square Scores: r_score_linear = 0.9230113113764172 r_score_Poly = -0.054182215904108544 r_score_RBF = -0.05740178748055347 r_score_Sigmoid = -0.0574927436944947
For C =0.05, R_Square Scores: r_score_linear = 0.9390871668294954 r_score_Poly = -0.057155343990719754 r_score_RBF = -0.05747783897509007 r_score_Sigmoid = -0.05748693570044017	For C =1.0, R_Square Scores: r_score_linear = 0.8950779234313312 r_score_Poly = -0.050890117824376135 r_score_RBF = -0.05731730927224388 r_score_Sigmoid = -0.0574991971677592
For C =0.1, R_Square Scores: r_score_linear = 0.9375216515526494 r_score_Poly = -0.056824517369874705 r_score_RBF = -0.057469387821565965 r_score_Sigmoid = -0.05748758102694351	For C =1.5, R_Square Scores: r_score_linear = 0.8985321607927548 r_score_Poly = -0.047061369861496116 r_score_RBF = -0.057232855750551925 r_score_Sigmoid = -0.057505650795273144

**Best R2 Score for the parameters are kernel = linear & C =0.05 and
it's value is: 0.9390871668294954**

Predicted Result of the chosen model is : 44394.75

3.Decision Regression Regression:

Regressor :

Parameters Tweaked are criterion , splitter and max_features

criterion values are : "squared_error", "absolute_error" & "poisson"

splitter values are : "best"& "random"

max_features values are : "sqrt" , "log2", 0.5, 0.6, 0.7, 0.8 & 0.9

R2 Score :

1. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :sqrt. are : 0.49376528621808335
2. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :log2. are : 0.78595656172054
3. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :0.5. are : 0.7637520767864423
4. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :0.6. are : 0.9481841681231585
5. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :0.7. are : 0.756948682694516
6. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :0.8. are : 0.7423000077754102
7. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :0.9. are : 0.7356247083987663
8. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :sqrt. are : 0.4846348792178714
9. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :log2. are : 0.17483647135007752
10. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :0.5. are : 0.6164176379856147
11. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :0.6. are : 0.4382606271202928
12. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :0.7. are : 0.8936504026266254
13. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :0.8. are : 0.9070427610263896
14. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :0.9. are : 0.8919637817846409
15. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :sqrt. are : 0.6865758543325604
16. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :log2. are : 0.7930316926560225
17. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :0.5. are : 0.910639247883286
18. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :0.6. are : 0.6774737572434544
19. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :0.7. are : 0.508593963491398
20. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :0.8. are : 0.9164781322528359
21. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :0.9. are : 0.9678001862942439
22. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :sqrt. are : -0.24152848626345103
23. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :log2. are : 0.7268601734415707
24. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :0.5. are : -0.4775162236762964
25. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :0.6. are : 0.679226147313505
26. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :0.7. are : 0.6761541108486653
27. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :0.8. are : 0.621517582049308
28. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :0.9. are : 0.6560336785877622
29. The result of the parametric combinations OF criterion :poisson splitter :best max_features :sqrt. are : 0.5921864421236449
30. The result of the parametric combinations OF criterion :poisson splitter :best max_features :log2. are : -0.156699803909512
31. The result of the parametric combinations OF criterion :poisson splitter :best max_features :0.5. are : 0.741893064383379
32. The result of the parametric combinations OF criterion :poisson splitter :best max_features :0.6. are : 0.7496175062770543
33. The result of the parametric combinations OF criterion :poisson splitter :best max_features :0.7. are : 0.9213283445240934
34. The result of the parametric combinations OF criterion :poisson splitter :best max_features :0.8. are : 0.9271806456008901
35. The result of the parametric combinations OF criterion :poisson splitter :best max_features :0.9. are : 0.9067202900372074
36. The result of the parametric combinations OF criterion :poisson splitter :random max_features :sqrt. are : 0.006862884443456285
37. The result of the parametric combinations OF criterion :poisson splitter :random max_features :log2. are : 0.8720388479999563
38. The result of the parametric combinations OF criterion :poisson splitter :random max_features :0.5. are : 0.6140876448970132
39. The result of the parametric combinations OF criterion :poisson splitter :random max_features :0.6. are : 0.8954843806988316
40. The result of the parametric combinations OF criterion :poisson splitter :random max_features :0.7. are : 0.612430957222329
41. The result of the parametric combinations OF criterion :poisson splitter :random max_features :0.8. are : 0.879522055117929
42. The result of the parametric combinations OF criterion :poisson splitter :random max_features :0.9. are : 0.6926459670740703

The max value among the Result is S.No :21 Which is 0.9678001862942439

Best R2 Score for the parameters are criterion =absolute_error , splitter = best & max_features : 0.9 and it's value is: 0.9678001862942439

Predicted Result of the chosen model is : 49490.75